MILITARY
SPECIFICATION

MIL-S-11779(SigC)
21 February 1952
Superseding U.S. Army
Specification 74-30-B
(In part) 11 January 1944

SCALE, METEOROLOGICAL, ML-125( ), ML-177( ), AND ML-301( )/TM

1. SCOPE

1.1 This specification covers the following meteorological scales:

Scale ML-125( )
Scale ML-177( )
Scale ML-301( )/TM

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-B-36	Bags, Paper, Grocers'
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)

MILITARY

JAN-P-105	Packaging and Packing for Overseas Shipment-Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment-Boxes; Wood, Nailed
JAN-P-116	Packaging and Packing for Overseas Shipment-Preservation, Methods of
JAN-P-120	Packaging and Packing for Overseas Shipment-Cartons, Folding, Paperboard
JAN-P-133	Packaging and Packing for Overseas Shipment-Boxes, Set-Up, Paperboard
MIL-I-10184	Inspection Procedures, General; for Signal Corps Equipment
MIL-L-10547	Liners, Case, Waterproof

U. S. NAVY

33-P-39	Plastic, Transparent, Sheet, Vinyl Chloride-Vinyl Acetate Copolymer

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments

DRAWINGS

SIGNAL CORPS

SC-D-5002	Scales ML-125( ), ML-177( ), and ML-301( )/TM

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Preproduction samples.- Requirements for preproduction samples shall be as specified in Specification MIL-I-10184.

3.1.1 Quantities.- After award of contract, three preproduction samples of each type scale on order, shall be furnished.

3.2 Master plates.- Master plates, of each type of scale on order will be supplied to the contractor upon his application to the contracting officer.

3.3 Construction.- Each scale shall be constructed in accordance with Drawing SC-D-5002 and the requirements herein.

3.3.1 Assembly.- The scale shall be assembled from two sheets of unequal thickness. The thinner sheet shall form the lower lamination when the scale is placed on a horizontal surface with the printing appearing, from above, as it is shown on the drawing. To afford proper protection, the data shall be printed on either one (at the contractor's option) of the two surfaces which are laminated together.

3.3.2 Plastic sheets.- The sheets used for making the scales shall be of a clear, transparent, hard, flexible, waterproof plastic which meets the requirements for Type A, Class a material in accordance with Specification 33-P-39.

3.3.3 Printing.- All printing shall be black, dense, completely opaque and evenly pigmented. All lines shall be straight, narrow and sharply defined. There shall be not appreciable widening of the lines caused by the process of printing or laminating. The edges of all lines, figures, and letters shall be sharp and clean-cut. All graduation lines shall be of uniform width to the edge of the scale.

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3.3.4 Punching.- The holes for the protractors in Scales ML-125( ) and ML-301( )/TM shall be punches in the locations shown on Drawing SC-D-5002. The centers of these holes are indicated on the master plates by black dots placed on the circumference lines of the protractors. The edges of the holes shall be free from burrs after punching. There shall be no trace of opaque areas around the holes or of separation of the laminated sheets. On visual inspection, the holes shall appear to be accurately centered half-way between the radial lines of the protractor except that the three holes on the bottom line of the protractor shall be centered on that line. A radial deviation of 1/64 inch in centering will be acceptable.

3.3.5 Appearance of finished scales.- The finished scale shall be as nearly colorless as possible. Any small amount of coloring present shall be evenly distributed throughout the scale. The scale shall be free from bubbles and opaque or discolored spots. The surfaces of the finished scale shall be smooth, glossy and free from scratches, rough spots, wrinkles, and wavy areas. Hair-line scratches and other minute blemishes which may be observed when looking through the scale at a strong light are permissible if they cannot be seen when the scale is laid on a white mat surface.

3.4 Flatness.- The scale, when laid on a flat surface, shall show no warpage greater than 1/16 inch at any point and the average shall be 1/32 inch or less. This requirement shall apply both before and after the scale is subjected to the test of 4.3.

3.5 Workmanship.- The scales shall be manufactured in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Acceptance inspection.- Acceptance inspection shall conform to Specification MIL-I-10184 and Table I.

Table I. Inspection procedures

Inspection or test	Rqts. par..	Test par.	Preproduction	Acceptance inspection (Production inspection)	AQL
Visual inspection	3.3 3.3.3 3.3.4 3.3.5 3.5	4.2	X	X	
Heat and humidity test	3.4	4.3	X		

NOTES: (a) Other preproduction tests may be performed to determine conformance with specification requirements as follows: (1) Nondestructive tests on preproduction samples of the scales, and (2) Destructive or nondestructive tests on separate preproduction samples of parts or materials.

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4.2 Visual inspection.- The scales shall be visually inspected for compliance with the requirements of this specification and the drawing as to workmanship, transparency, thickness, smoothness of surface, seperation of sheets, centering of punches holes, printing, etc.

4.3 Heat and humidity test.- The scales shall be exposed, in an air circulating oven, to a temperature of 130F at 90 percent relative humidity for 24 hours. After the exposure, the plastic scales shall be placed on a flat surface. When in this position, no portion of the surface shall be more than 1/16 inch above the surface on which it is resting. In addition, there shall be no sign of discoloration or separation of the laminations.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Scales ML-125( ), ML-177( ), and ML-301( )/TM shall be preserved and packaged in accordance with best commercial practice. The equipment shall be adequately cushioned, blocked, and braced, using suitable materials and containers as required, in such a manner as to afford maximum protection from hazards normally encountered in storage and transit.

5.1.2 For overseas shipment.- Preserve and package meteorological scales for mechanical and physical protection only, in accordance with procedures specified for the designated methods, as prescribed in Specification JAN-P-116, and further described in the following paragraphs.

5.1.2.1 Meteorological scales.- Cushion each meteorological scale by inclosing each scale between non-bending chipboard pads. The pads shall extend a minimum of 1/8 inch beyond the edges of each scale. Secure the pads together with gummed kraft tape, complying with Specification UU-T-111. Inclose each cushioned scale within a close fitting bag, fabricated of 60 lbs. basic weight kraft paper. Bags shall comply with Specification UU-B-36. Triple fold closure and staple through the triple fold.

5.1.2.2 Intermediate package.- Place ten (10) packaged scales within a close fitting folding carton or set-up box complying with Specification JAN-P-120 or JAN-P-133, respectively. Seal entire closure with gummed kraft tape, complying with Specification UU-T-111. All packages within one intermediate container shall bear the same stock number.

5.2 Packing.-

5.2.1 For domestic shipment.- Meteorological scales shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose in such a manner as to afford maximum protection from normal hazards of handling and transportation, and to insure acceptance at and safe delivery to the designated point by common or other carriers, at the lowest applicable rate.

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5.2.2 For overseas shipment.- Place a suitable quantity of meteorological scales packaged as specified in 5.1.2 through 5.1.2.2 within a wood cleated plywood box or nailed wood box, complying with Specification JAN-P-105 or JAN-P-106, respectively. Fabricate the boxes to fit the contents snugly. Each shipping container shall be fitted with a waterproof case liner, complying with Specification MIL-L-10547, and the case liner shall be closed and sealed as specified in Appendix I, of the referenced specification. Close and strap the shipping containers as specified in the appendix of the applicable box specification. The maximum gross weight of any shipping container shall not exceed approximately 150 pounds.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129. On overseas shipping containers, the Signal Corps Insignia shall be placed on the lower left hand corner of the side containing contact data.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type required.

(c) Whether domestic or overseas shipment is required.

(d) Samples required. (See 3.1)

(e) Marking and shipping of samples

6.2 Preparation for delivery.- Procurement document shall state whether domestic or overseas packing and packaging is required.

6.3 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example AN/PIQ-1W. As soon as possible after the award of the contract, the contractor should apply to the contracting officer for such information. (See 1.1)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in anyway supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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